I、Laplace. $\Delta = \sum \dfrac{\partial^2}{\partial x_i^2}$. 对 n 维. $\Delta(|r|^{2-n}) = 0$.

1、对 $E(r)$. $\Delta = \dfrac{d^2}{dr^2} + \dfrac{n-1}{r} \cdot \dfrac{d}{dr}$. 令 $W(r) = \dfrac{dE}{dr}$.

$r^{n-1}\dfrac{dW}{dr} + (n-1)r^{n-2}W$ 于是有：$(r^{n-1}W)' = 0 \Rightarrow W = C \cdot r^{1-n}$.

$\Rightarrow E = \begin{cases} C_1 \ln \dfrac{1}{r} & n=2 \\ C_2 r^{2-n} & n \geq 3 \end{cases} \quad \Rightarrow \quad E = \begin{cases} -\dfrac{1}{2\pi} \ln \dfrac{1}{r} & n=2 \\ \dfrac{r^{2-n}}{(2-n)|S^{n-1}|} & n \geq 3 \end{cases}$

2、

$$\iiint_\Omega (u\Delta v - v\Delta u)dx = \iint_{\partial\Omega} (u\frac{\partial v}{\partial n} - v\frac{\partial u}{\partial n})ds.$$

$$\iint_{\partial\Omega} u\frac{\partial v}{\partial n}ds = \iiint_\Omega u\Delta v dx + \iiint_\Omega \sum \frac{\partial u}{\partial x_i} \cdot \frac{\partial v}{\partial x_i}dx.$$

3、

$$u(x) = \frac{1}{4\pi}\iint_{\partial\Omega} \left(\frac{1}{r}\frac{\partial u}{\partial n} - u\frac{\partial}{\partial n}(\frac{1}{r})\right)ds - \frac{1}{4\pi}\iiint_\Omega \frac{\Delta u}{r}dx$$

取 $\varphi \in C_0^\infty(\Omega)$. 在 $\partial\Omega$, $\varphi = 0$。于是 $C_n = -\dfrac{1}{4\pi}$

$$\varphi(Q) = \int \delta(x-y)\varphi(y)dy = \langle \Delta(C_n r^{2-n}), \varphi \rangle = C_n \iiint \frac{\Delta\varphi}{r}dy = -\frac{1}{4\pi}\iiint \frac{\Delta\varphi}{r}dx$$

4、取 $v = 1$. 平均值公式. $u(x) = \dfrac{1}{4\pi R^2}\iint_{\partial B(x,R)} u(y)dS$.

再在两边对 R 积分得体平均值. $u(x) = (\dfrac{4\pi r^3}{3})^{-1}\iiint_{B(x,r)} u(y)dy$.

5、极值原理. Dirichlet 问题. 解的唯一性, 稳定性.

6、e.g. $\begin{cases} \Delta u = 0 \\ \frac{\partial u}{\partial n}|_{\partial\Omega} = f \end{cases}$. 若 u 为解, 则所有解为 $u + C$. 在 Green 第一公式中取 $v = u$ 即可.

7、$G(P,Q) = -\dfrac{1}{4\pi}\left(\dfrac{1}{r(P,Q)} - v(P,Q)\right)$, 称为 Green 函数，调和，P = Q . 求解 Dirichlet 问题关键在于求 G，在已经知道基本解的情况下关键在于构造调和函数 v。一般地，有：

$$G(P,Q) = \Gamma(P,Q) + v(P,Q)$$

$$u(Q) = \iint\limits_{\partial\Omega} g(P)\frac{\partial}{\partial n}G(P,Q)\mathrm{d}S_P \quad \Delta G(P,Q) = \Delta\left(\Gamma(P,Q) + v(P,Q)\right) = \delta(P,Q)$$

8、交换性：$G(Q_1,Q_2) = G(Q_2,Q_1)$。挖两个球，用格林公式。$u = G(P,Q_2)$ $v = G(P,Q_1)$。

$$\iint\limits_{\partial B_\varepsilon(Q_1)+\partial B_\varepsilon(Q_2)} G(P,Q_1)\frac{\partial G(P,Q_2)}{\partial n}\mathrm{d}S = \iint\limits_{\partial B_\varepsilon(Q_1)+\partial B_\varepsilon(Q_2)} G(P,Q_2)\frac{\partial G(P,Q_1)}{\partial n}\mathrm{d}S$$

9、半平面的 green 函数：设 $Q(x,y)$，$Q_1(x,-y)$，则：

$$G(P,Q) = \Gamma(P,Q) + v(P) = -\frac{1}{2\pi}\ln\frac{1}{r(P,Q)} + \frac{1}{2\pi}\ln\frac{1}{r(P,Q_1)}$$

$$\frac{\partial G(P,Q)}{\partial n} = -\frac{1}{2\pi}\left[\frac{\partial}{\partial y_P}\ln r - \frac{\partial}{\partial y_P}\ln r_1\right] = -\frac{1}{2\pi}\left[\frac{y_P - y}{r^2} - \frac{y + y_P}{r_1^2}\right]$$

$$u(x,y)=\frac{1}{\pi}\int_{-\infty}^{+\infty}f(x_P)\frac{y}{(x-x_P)^2+y^2}\mathrm{d}x_P$$

10、圆的 green 函数：由相似：$rR=r_1\rho$。$\frac{\partial}{\partial n}=\frac{\partial}{\partial \rho_P}$。

$$G(P,Q)=-\frac{1}{2\pi}\left[\ln\frac{1}{r}-\ln\frac{R}{\rho r_1}\right],\quad \frac{\partial}{\partial n}G(P,Q)=\frac{1}{2\pi R}\frac{R^2-\rho^2}{r^2}\frac{\rho_P}{R}$$

$$u(\rho,\theta)=\frac{1}{2\pi}\int_0^{2\pi}f(\varphi)\frac{R^2-\rho^2}{R^2-2R\rho\cos(\theta-\varphi)+\rho^2}\mathrm{d}\varphi$$

11、如果满足平均值公式，则调和。（Dirichlet 问题加极值原理，最后由 B 的任意性。）

12、可去奇点定理。u(Q) 在 A 附近调和，且有 $\lim\limits_{Q\to A}\frac{u(Q)}{\ln r(A,Q)}=0\quad(n=2)$。则可加定义使在整个区域调和。(n 不等于 2 时，分母为 r^{2-n})

在 $B\equiv B_R(A)$ 中考虑即可。u_1 为 dirichlet 问题解，$w=u-u_1$。想让 w 在 $B-A$ 内恒为 0. 用极值原理使其在环上被 v_ϵ 控制，再令 $\epsilon\to 0$ 完成证明。

$$v_\varepsilon(Q)=\varepsilon[\ln\frac{1}{r(A,Q)}-\ln\frac{1}{R}]\quad \lim_{Q\to A}\frac{|u|}{V_\epsilon}=0$$

13、刘维尔定理。考虑分别记以 0 和 x 为心、R 为半径的球为 $B_R(0)$, $B_R(x).|R|>|x|.|B_n|=C_nR^n$。

$$\begin{aligned}|u(x)-u(0)|&=|B_n|^{-1}\left|\int_{B_R(x)}u(y)dy-\int_{B_R(0)}u(y)dy\right|\leqslant|B_n|^{-1}\int_{B_R(x)\Delta B_R(0)}|u(y)|dy.\\&\leqslant||u||_\infty|B_n|^{-1}\left(\int_{|y|<R,|y-x|>R}dy+\int_{|y|>R,|y-x|<R}dy\right)\\&\leqslant||u||_\infty|B_n|^{-1}\int_{R-|x|<|y|<R+|x|}dy=O(1/R)\end{aligned}$$

14、Harnack 定理 $\{u_n(x)\}$ 为 Ω 中调和序列，且 $u_n(x)\rightrightarrows u(x)$，则 u 也调和。只需对每个点证明满足平均值公式即可。

15、设 u 在 $B_R(0)$ 内非负调和, $u\in C(\overline{B_R})$, 对 $\forall P\in B_R(0)$ 有 $\frac{R-\rho}{R+\rho}u(Q)\leqslant u(P)\leqslant\frac{R+\rho}{R-\rho}u(Q)$。

由泊松积分公式，再注意 $\cos(\theta-\varphi)\in[-1,1]$, 得:

$$\frac{R^2-\rho^2}{(R+\rho)^2}\leqslant\frac{R^2-\rho^2}{R^2+\rho^2-2R\rho\cos(\theta-\varphi)}\leqslant\frac{R^2-\rho^2}{(R-\rho)^2}$$

。

II、抛物方程，热传导方程，记算子为 ∂_R

1、基本解：

$$\frac{\partial}{\partial t}\hat{E}(\xi,t)+a^2|\xi|^2\hat{E}(\xi,t)=\delta(t)\Rightarrow\hat{E}(\xi,t)=H(t)\cdot e^{-a^2|\xi|^2 t}$$

由 $F^{-1}\left(e^{-\frac{|\xi|^2}{2}}\right)=(2\pi)^{-\frac{n}{2}}e^{-\frac{|x|^2}{2}}$，得 $E(x,t)=F^{-1}\left[H(t)e^{-a^2|\xi|^2 t}\right]=H(t)\left(4\pi a^2 t\right)^{-\frac{n}{2}}e^{-\frac{|x|^2}{4a^2 t}},t>0$

2、柯西问题：P1 傅里叶变换，P2duhamel。或基本解。

$$(P)\begin{cases}\dfrac{\partial u}{\partial t}-a^2\Delta u=f(x,t)\\ u|_{t=0}=\varphi(x)\end{cases}\xrightarrow{\text{叠加原理}}(P_1)\begin{cases}\dfrac{\partial u}{\partial t}-a^2\Delta u=0\\ u|_{t=0}=\varphi(x)\end{cases}\quad(P_2)\begin{cases}\dfrac{\partial u}{\partial t}-a^2\Delta u=f(x,t)\\ u|_{t=0}=0\end{cases}$$

(P1): $$\begin{cases}\dfrac{d\widehat{u}(\xi,t)}{dt}+a^2|\xi|^2\widehat{u}(\xi,t)=0\\ \widehat{u}(\xi,t)|_{t=0}=\widehat{\varphi}(\xi)\end{cases}\Rightarrow\widehat{u}(\xi,t)=\widehat{\varphi}(\xi)e^{-a^2|\xi|^2 t},\quad t\geq 0\Rightarrow u(x,t)=\mathcal{F}^{-1}(\widehat{u}(\xi,t))$$

$$\hat{f}_2=e^{-a^2|\xi|^2 t}\quad\Rightarrow f_2=(4\pi a^2 t)^{-\frac{n}{2}}e^{-\frac{|x|^2}{4a^2 t}}=(4\pi a^2 t)^{-\frac{n}{2}}e^{-\frac{|x|^2}{4a^2 t}}$$

$$\Rightarrow u_1(x,t)=\mathcal{F}^{-1}(\hat{\varphi}(\xi)\cdot e^{-a^2|\xi|^2 t})=E(x,t)\underset{x}{*}\varphi(x)=(4\pi a^2 t)^{-\frac{n}{2}}\int_{\mathbb{R}^n}\varphi(y)e^{-\frac{|x-y|^2}{4a^2 t}}dy$$

(P2): 令 U 满足 $\begin{cases}\left(\dfrac{\partial}{\partial t}-a^2\Delta\right)U=0, & t>\tau\\ U(x,t,\tau)|_{t=\tau}=f(x,\tau)\end{cases}$，令$u(x,t)=\int_0^t U(x,t;\tau)\,d\tau$ 即为解。

于是 $U(x,t,\tau)=E(x,t-\tau)*_{(x)}f(x,\tau)=\left(4\pi a^2(t-\tau)\right)^{-\frac{n}{2}}\int_{\mathbb{R}^n}f(y,\tau)e^{-\frac{|x-y|^2}{4a^2(t-\tau)}}dy$

$$\begin{aligned}u_2(x,t)=\int_0^t U(x,t,\tau)d\tau&=\int_0^t\left(4\pi a^2(t-\tau)\right)^{-\frac{n}{2}}\int_{\mathbb{R}^n}f(y,\tau)\exp\left(-\frac{|x-y|^2}{4a^2(t-\tau)}\right)dyd\tau\\&=\int_0^t\int_{\mathbb{R}^n}f(y,\tau)\left(4\pi a^2(t-\tau)\right)^{-\frac{n}{2}}\exp\left(-\frac{|x-y|^2}{4a^2(t-\tau)}\right)dyd\tau\end{aligned}$$

2′、用基本解，设 $\tilde{u}(x,t)=H(t)u(x,t)$。

$$\left(\frac{\partial}{\partial t}-a^2\Delta\right)\tilde{u}(x,t)=\left(\frac{\partial}{\partial t}-a^2\Delta\right)(H(t)u(x,t))\triangleq F(x,t)\Rightarrow\tilde{u}(x,t)=E(x,t)*F(x,t)$$

$$\left(\frac{\partial}{\partial t}-a^2\Delta\right)(H(t)u(x,t))=\delta(t)u(x,t)+H(t)\frac{\partial}{\partial t}u(x,t)-a^2H(t)\Delta u(x,t)=\delta(t)u(x,t)+H(t)f(x,t)$$

则 $F(x,t)=\delta(t)\varphi(x)+H(t)f(x,t)\quad\tilde{u}(t,x)=E(x,t)*F(x,t)=E(x,t)*(\delta(t)\varphi(x)+H(t)f(x,t))$。
对于前一项，考虑 $\delta_\varepsilon(t)=\delta(t)*\Phi_\varepsilon(t)\in C_0^\infty$，

$$(\delta_\varepsilon(t)\varphi(x))*E(x,t)=\int_{\mathbb{R}}\int_{\mathbb{R}^n}\delta_\varepsilon(\tau)\varphi(y)E(x-y,t-\tau)\mathrm{d}\tau\mathrm{d}y=\big\langle\varphi(y),\langle\delta_\varepsilon(\tau),E(x-y,t-\tau)\rangle\big\rangle$$

两边取极限 $(\delta(t)\cdot\varphi(x))*E(x,t)=\big\langle\varphi(y),E(x-y,t)\big\rangle=\ E(x,t)\underset{(x)}{*}\varphi(x)$。
对于第二项只能把 heaviside 函数化简成 τ 的取值范围，形式没有本质区别。

3、极值原理，设在矩形内部 $\dfrac{\partial u}{\partial t}-a^2\Delta u=0$，则再 Γ_T（侧＋底）取最大最小值。

设 u 全局最大值 M，边界最大值 m，在内部 (x_0,t_0) 取最大值。$v(x,t)=u(x,t)+\dfrac{M-m}{4l^2}(x-x_0)^2$。
v 在内部 (x_1,t_1) 取最大值。由计算，$\partial_R v<0$。

$$\left.\frac{\partial^2 v}{\partial x^2}\right|_{(x_1,t_1)}\leq 0,\left.\frac{\partial v}{\partial t}\right|_{(x_1,t_1)}\geq 0\Rightarrow\frac{\partial v}{\partial t}-a^2\frac{\partial^2 v}{\partial x^2}\geq 0\quad(x_1,t_1)$$

4、边初值问题（除 ∂_R 外还在两个侧边和底边有条件。）解的唯一性和稳定性。

5、有界函数类中 Cauchy 问题的唯一性和稳定性。$\begin{cases}(\partial_t - a^2\partial_x^2)u = 0, & t>0\\ u|_{t=0} = \varphi(x)\end{cases}$。

唯一性：$u = u_1 - u_2$，对于固定的 (x_0, t_0) 证明 $u(x_0, t_0) = 0$。首先人为划定矩形区域 $R_0 = \{(x,t) \mid |x-x_0| \leqslant L, 0 \leqslant t \leqslant t_0\}$ 用边初值问题的极值原理，再令 v 中的 L 趋于无穷即得。注意验证 v 满足边初值条件。

$$v(x,t) = \frac{4B}{L^2}\left(\frac{(x-x_0)^2}{2} + a^2 t\right) \in C(R_0).$$

稳定性：$|\varphi_1(x) - \varphi_2(x)| \leqslant \eta$，考虑 $v(x,t) = \dfrac{4B}{L^2}\left(\dfrac{(x-x_0)^2}{2} + a^2 t\right) + \eta$ 即可。

6、解边初值问题。首先化为齐次问题。$v(x,t) = \mu_1(t) + \dfrac{x}{\ell}(\mu_2(t) - \mu_1(t))$。再叠加原理。

$$(P)\begin{cases}\dfrac{\partial u}{\partial t} - a^2\dfrac{\partial^2 u}{\partial x^2} = f(x,t), t>0\\ u(0,t) = \mu_1(t), \quad u(\ell,t) = \mu_2(t)\\ u(x,0) = \varphi(x)\end{cases} \xrightarrow{\bar u = u - v} \begin{cases}\dfrac{\partial \bar u}{\partial t} - a^2\dfrac{\partial^2 \bar u}{\partial x^2} = f(x,t) - \left(\dfrac{\partial v}{\partial t} - a^2\dfrac{\partial^2 v}{\partial x^2}\right) \triangleq \bar f(x,t)\\ \bar u(0,t) = \bar u(\ell,t) = 0\\ \bar u(x,0) = \varphi(x) - v(x,0) \triangleq \bar\varphi(x)\end{cases}$$

然后对右边用叠加原理。P2duhamel 原理。

$$(P_1)\begin{cases}\dfrac{\partial u}{\partial t} - a^2\dfrac{\partial^2 u}{\partial x^2} = 0, \quad t>0\\ u(0,t) = u(\ell,t) = 0\\ u|_{t=0} = \varphi(x)\end{cases} \qquad (P_2)\begin{cases}\dfrac{\partial u}{\partial t} - a^2\dfrac{\partial^2 u}{\partial x^2} = f(x,t)\\ u(0,t) = u(\ell,t) = 0\\ u|_{t=0} = 0\end{cases}$$

7、对（P1）Fourier 方法（分离变量法）。设 $u(x,t) = X(x)T(t)$，我们要求 X，T 不恒为 0.

$$X(x)T'(t) - a^2 X''(x)T(t) = 0 \quad \Rightarrow \quad \frac{X''(x)}{X(x)} = \frac{T'(t)}{a^2 T(t)} \triangleq -\mu$$

$\Rightarrow X''(x) + \mu X(x) = 0,\ X(0) = X(\ell) = 0$

$\mu < 0$ 时 $X(x) = c_1 e^{\sqrt{-\mu}x} + c_2 e^{-\sqrt{-\mu}x}$, $\mu = 0$ 时 $X''(x) = 0 \quad X(x) = c_1 x + c_2$.

$\mu > 0$ 时 $X(x) = c_1\cos\sqrt{\mu}x + c_2\sin\sqrt{\mu}x$,$\mu = \left(\dfrac{k\pi}{\ell}\right)^2$ $\forall c_2$ 对 $k = 0, \pm1, \pm2, \dots X_k(x) = C_k \sin\dfrac{k\pi}{\ell}x \quad k = 1, 2, \dots$。于是 $T_k'(t) + a^2\mu_k T_k(t) = 0 \quad \Rightarrow T_k(t) = \overline{A_k} e^{-a^2\left(\frac{k\pi}{l}\right)^2 t} \quad k = 1, 2, \cdots$，从而：

$$u_k(x,t) = X_k(x)T_k(t) = A_k e^{-a^2\left(\frac{k\pi}{l}\right)^2 t}\sin\frac{k\pi}{l}x, \quad u(x,t) = \sum^{\infty} u_k(x,t)$$

考虑用傅里叶系数求 A_k，$\varphi(x) = \sum_{k=1}^{\infty} A_k \sin\dfrac{k\pi x}{l} \Rightarrow A_k = \dfrac{2}{l}\displaystyle\int_0^l \varphi(x)\sin\dfrac{k\pi x}{l}\,dx$

8、正交性：$\displaystyle\int_0^l X_m(x)X_n(x)\,dx = 0$。对下式右边两边积分＋分部积分即得。

$$X_n\left(X_m'' + \mu_m X_m\right) - X_m\left(X_n'' + \mu_n X_n\right) = 0 \Rightarrow (\mu_m - \mu_n)X_m X_n = X_n'' X_m - X_m'' X_n$$

9、对（P2），首先看 duhamel 原理。若 $w(x,t,\tau)$ 是左边初边值问题的解，则 $u(x,t) = \displaystyle\int_0^t w(x,t,\tau)\,d\tau$

是 (P2) 的解。

$$\begin{cases} \dfrac{\partial w}{\partial t} - a^2 \dfrac{\partial^2 w}{\partial x^2} = 0 \quad t > \tau \\ w(x,\tau,\tau) = f(x,\tau) \\ w(0,t,\tau) = w(l,t,\tau) = 0 \end{cases} \qquad (P2) \begin{cases} \dfrac{\partial u}{\partial t} - a^2 \dfrac{\partial^2 u}{\partial x^2} = f(x,t) \quad t > 0 \\ u(0,t) = u(l,t) = 0 \\ u|_{t=0} = 0 \end{cases}$$

从而 $w(x,t,\tau) = \sum_{k=1}^{\infty} A_k e^{-a^2\left(\frac{k\pi}{l}\right)^2(t-\tau)} \sin\frac{k\pi}{l}x$，$u_2(x,t) = \int_0^t \left(\sum_{k=1}^{\infty} A_k e^{-a^2\left(\frac{k\pi}{l}\right)^2(t-\tau)} \sin\frac{k\pi}{l}x \right) d\tau$

于是，(P) 的解即为 $u_1 + u_2 + v$。

10、还可以考虑常数变易法。$\left\{X_k(x) = \sin\frac{k\pi}{l}x\right\}$ 是完全正交系。用 X_k 把 u, f, ϕ 展开。然后常数变易。$u(x,t) = \sum_{k=1}^{\infty} u_k(t)\sin\frac{k\pi}{l}x, f(x,t) = \sum_k f_k(t)\sin\frac{k\pi}{l}x, \varphi(x) = \sum_k \varphi_k \sin\frac{k\pi}{l}x.$ 代回原方程组。

$$\begin{cases} u_k'(t)\sin\frac{k\pi}{l}x + a^2\left(\frac{k\pi}{l}\right)^2 u_k(t)\sin\frac{k\pi}{l}x = f_k(t)\sin\frac{k\pi}{l}x \\ u_k(0)\sin\frac{k\pi}{l}x = \varphi_k\sin\frac{k\pi}{l}x \end{cases} \Rightarrow \begin{cases} u_k'(t) + a^2\left(\frac{k\pi}{l}\right)^2 u_k(t) = f_k(t) \\ u_k(0) = \varphi_k \end{cases}$$

$u_k(t) = \left(\varphi_k + \int_0^t f_k(s) e^{a^2\left(\frac{k\pi}{l}\right)^2 s} ds\right) e^{-a^2\left(\frac{k\pi}{l}\right)^2 t}$ 于是 $u(x,t) = \sum_{k=1}^{\infty} u_k(t)\sin\frac{k\pi}{l}x$，$u(x,t) + v(x,t)$ 为解。

11、解 $\begin{cases} \dfrac{\partial u}{\partial t} - a^2\dfrac{\partial^2 u}{\partial x^2} = x(l-x) \quad 0 < x < l \quad t > 0 \\ u(0,t) = 0 \qquad \dfrac{\partial u}{\partial x}(l,t) = 1 \\ u|_{t=0} = \sin\dfrac{\pi x}{l} \end{cases}$.

$v(x,t) = x$，$X_k(x) = C_k \sin\frac{(k+\frac{1}{2})\pi}{l}x \quad k = 0, 1, \cdots \quad C_k$ 任意常数，$T_k(t) = \overline{A_k} e^{-a^2\left(\frac{(k+\frac{1}{2})\pi}{l}\right)^2 t}$。

$$u(x,t) = \sum_{k=0}^{\infty} u_k(x,t) = \sum_{k=0}^{\infty} A_k e^{-a^2\left(\frac{(k+\frac{1}{2})\pi}{l}\right)^2 t} \sin\frac{k\pi + \frac{\pi}{2}}{l}x$$

。再用 duhamel 原理，或者利用展开：

$$\begin{cases} x(l-x) = \sum_{k=0}^{\infty} A_k \sin\frac{k\pi+\frac{\pi}{2}}{l}x \\ \sin\frac{\pi x}{l} - x = \sum_0 B_k \sin\frac{k\pi+\frac{\pi}{2}}{l}x \\ u_1(x,t) = \sum_{k=0}^{\infty} u_k(t)\sin\frac{k\pi+\frac{\pi}{2}}{l}x \end{cases} \Rightarrow \begin{cases} u_k'(t) + a^2\left(\frac{k\pi+\frac{\pi}{2}}{l}\right)^2 u_k(t) = A_k \\ u_k(0) = B_k \end{cases}$$

$$u_k(t) = \left(B_k + \int_0^t A_k e^{a^2\left(\frac{k\pi+\frac{\pi}{2}}{l}\right)^2 s} ds\right) e^{-a^2\left(\frac{k\pi+\frac{\pi}{2}}{l}\right)^2 t} \Rightarrow u_1(x,t) = \sum_{k=0}^{\infty} u_k(t)\sin\frac{k\pi+\frac{\pi}{2}}{l}x$$

最后 $u(x,t) = u_1(x,t) + x$。

III、波动方程，弦振动方程。

1、为了做傅里叶逆变换，我们先推广 δ 的定义。设曲面 $S=\{x\mid p(x)=0,\mathrm{d}p(x)\neq 0,p\in C^\infty\}$，定义 $\delta(p(x))\in\mathscr{D}'(\mathbb{R}^n):\forall\varphi\in C_0^\infty(\mathbb{R}^n),\langle\delta(p(x)),\varphi\rangle:=\displaystyle\int_{p(x)=0}\varphi(x)\mathrm{d}s$。若 $\{x\mid p(x)=0\}$ 为紧集，则 $\delta(p(x))\in\mathscr{E}'(\mathbb{R}^n)\subset\mathbb{D}'\Rightarrow\widehat{\delta(p(x))}=\langle\delta(p(x)),e^{-ix\xi}\rangle=\displaystyle\int_{p(x)=0}e^{-ix\xi}\mathrm{d}s$。以 $n=3$ 为例:

$$
\begin{aligned}
F(\delta(r-|x|))(\xi)&=\int_{|x|=r}e^{-ix\xi}\mathrm{d}S_x=\int_{|x|=r}e^{-i|x||\xi|\cos\theta}\mathrm{d}S_x=\int_0^{2\pi}\int_0^{\pi}e^{-i|x||\xi|\cos\theta}r^2\sin\theta\mathrm{d}\theta\mathrm{d}\varphi\\
&\xlongequal{\cos\theta=y}-2\pi r^2\int_1^{-1}e^{-ir|\xi|y}\mathrm{d}y=\frac{2\pi r^2}{r|\xi|}\int_{-1}^{1}(\cos(r|\xi|y)-i\sin(r|\xi|y))\,\mathrm{d}(r|\xi|y)\\
&=\frac{2\pi r}{|\xi|}\sin(r|\xi|y)\bigg|_{-1}^{1}=\frac{4\pi r}{|\xi|}\sin(r|\xi|)
\end{aligned}
$$

2、基本解。傅里叶变换得 $\dfrac{\partial^2}{\partial t^2}\widehat{E}(\xi,t)+a^2|\xi|^2\widehat{E}(\xi,t)=0(*)$，设:
$\widehat{E}(\xi,t)=k_1(\xi,t)\sin(a|\xi|t)+k_2(\xi,t)\cos(a|\xi|t)$，待定 $k_1(\xi,t),k_2(\xi,t)$。我们设 $k_1'\sin(a|\xi|t)+k_2'\cos(a|\xi|t)=0$，利用 $(*)$ 得到：$\begin{cases}k_1'\sin(a|\xi|t)+k_2'\cos(a|\xi|t)=0\\ \dfrac{\partial^2}{\partial t^2}\widehat{E}(\xi,t)+a^2|\xi|^2\widehat{E}(\xi,t)=k_1'(a|\xi|)\cos(a|\xi|t)-k_2'(a|\xi|)\sin(a|\xi|t)=\delta(t)\end{cases}$，
$\Rightarrow k_2'(a|\xi|)\left(\cos^2(a|\xi|t)+\sin^2(a|\xi|t)\right)=\delta(t)\sin(a|\xi|t)\Rightarrow k_2'(\xi,t)=0$ 只需设 $k_2(\xi,t)=0$ 即可。代回:
$\begin{cases}k_1'\sin(a|\xi|t)=0\\ k_1'(a|\xi|)\cos(a|\xi|t)=\delta(t)\end{cases}\Rightarrow k_1'=\dfrac{\delta(t)}{a|\xi|}\Rightarrow k_1(\xi,t)=\begin{cases}\dfrac{H(t)}{a|\xi|} & ,t>0\\ \dfrac{-H(-t)}{a|\xi|} & ,t<0\end{cases}$

$$
\widehat{E}(\xi,t)=\begin{cases}\widehat{E}_+(\xi,t)=\dfrac{H(t)}{a|\xi|}\sin(a|\xi|t) & ,t>0\\ \widehat{E}_-(\xi,t)=\dfrac{-H(-t)}{a|\xi|}\sin(a|\xi|t) & ,t<0\end{cases}
$$

$$
\Rightarrow\begin{cases}\text{对}t>0,\text{令}r=at(a>0),\widehat{E_+}(\xi,t)=\dfrac{H(t)}{4\pi a^2t}\mathcal{F}(\delta(at-|x|)) & :E_+(x,t)=H(t)\dfrac{1}{4\pi a^2t}\delta(at-|x|)\\ \text{对}t<0,\text{令}r=-at(a>0),\widehat{E_-}(\xi,t)=\dfrac{-H(-t)}{4\pi a^2t}\mathcal{F}(\delta(at+|x|)) & :E_-(x,t)=-H(-t)\dfrac{1}{4\pi a^2t}\delta(at+|x|)\end{cases}
$$

3、还可以直接进行傅里叶逆变换。记 $|\xi|=\rho$

$$
\begin{aligned}
E_+(x,t)&=F^{-1}\left(\widehat{E_+}(\xi,t)\right)=(2\pi)^{-3}H(t)\int_{\mathbb{R}^3}\frac{\sin(a|\xi|t)}{a|\xi|}e^{ix\xi}\mathrm{d}\xi\\
&=(2\pi)^{-3}H(t)\int_0^{+\infty}\frac{\sin(a\rho t)}{a\rho}\rho^2\mathrm{d}\rho\int_0^{2\pi}\mathrm{d}\varphi\int_0^{\pi}e^{i|x|\rho\cos\theta}\sin\theta\mathrm{d}\theta\\
&=\frac{H(t)}{a|x|}(2\pi)^{-2}\int_0^{+\infty}\sin(a\rho t)2\cdot\sin(|x|\rho)\mathrm{d}\rho=\frac{H(t)}{4\pi^2a|x|}\int_0^{+\infty}\cos(|x|-at)\rho-\cos(|x|+at)\rho\mathrm{d}\rho\\
&=\frac{H(t)}{4\pi^2a|x|}\lim_{A\to+\infty}\left(\frac{\sin[A(|x|-at)]}{|x|-at}-\frac{\sin[A(|x|+at)]}{|x|+at}\right)\\
&\xlongequal{\lim_{n\to\infty}\frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\sin nx}{x}\mathrm{d}x=\delta(x)}\frac{H(t)}{4\pi a|x|}(\delta(|x|-at)-\delta(|x|+at))=\frac{H(t)}{4\pi a|x|}\delta(|x|-at),t>0
\end{aligned}
$$

由 $\operatorname{supp}\delta=\{|x|=at\}\Rightarrow E_+(x,t)=H(t)\dfrac{1}{4\pi a^2t}\delta(at-|x|),t>0$。$t<0$ 同理。

4、一维波动 (弦振动) 方程的初值问题（Cauchy 问题）。(P) $\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = f(x,t) \quad (x\in\mathbb{R}) \quad t>0 \\ u|_{t=0} = \varphi(x) \\ \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = \psi(x) \end{cases}$ 。

$$(P_1) \quad \begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = 0 \quad t>0 \\ u|_{t=0} = \varphi(x) \\ \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = \psi(x) \end{cases} \qquad (P_2) \quad \begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = f(x,t) \quad t>0 \\ u|_{t=0} = 0 \\ \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = 0 \end{cases}$$

对于 (P_1) $\quad \xi = x - at \quad \eta = x + at$，于是 $\dfrac{\partial}{\partial\eta}\left(\dfrac{\partial u}{\partial\xi}\right) = 0$，积分得 $u = F(\xi) + G(\eta)$。由条件得：

$$\begin{cases} F(x) + G(x) = \varphi(x) \\ a(-F(x) + G(x)) + C = \displaystyle\int_{x_0}^{x} \psi(y)\,dy \end{cases} \qquad \begin{cases} F(x) = \dfrac{1}{2}\varphi(x) - \dfrac{1}{2a}\displaystyle\int_{x_0}^{x}\psi(y)\,dy + \dfrac{C}{2a} \\ G(x) = \dfrac{1}{2}\varphi(x) + \dfrac{1}{2a}\displaystyle\int_{x_0}^{x}\psi(y)\,dy - \dfrac{C}{2a} \end{cases}$$

$u(x,t) = \dfrac{1}{2}\varphi(x-at) + \dfrac{1}{2}\varphi(x+at) + \dfrac{1}{2a}\displaystyle\int_{x-at}^{x+at}\psi(y)\,dy$.(达朗贝尔等式)

对 (P2)，$w(x,t,\tau)$ 为齐次方程 $\begin{cases} \dfrac{\partial^2 w}{\partial t^2} - a^2\dfrac{\partial^2 w}{\partial x^2} = 0 \quad t>\tau \\ w|_{t=\tau} = 0 \\ \left.\dfrac{\partial w}{\partial t}\right|_{t=\tau} = f(x,\tau) \end{cases}$ 的解。$w(x,t,\tau) = \dfrac{1}{2a}\displaystyle\int_{x-a(t-\tau)}^{x+a(t-\tau)} f(y,\tau)\,dy$，

$u(x,t) = \displaystyle\int_0^t w(x,t,\tau)\,d\tau = \dfrac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)} f(y,\tau)\,dy\,d\tau$ 为 (P2) 的解.

5、一维波动（弦振动）方程的边初值问题。$\overline{u}(x,t) = \mu_1(t) + \dfrac{x}{l}(\mu_2(t) - \mu_1(t))$ 化零边值。

$$\text{原}\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = f(x,t) \\ u|_{t=0} = \varphi(x) \qquad \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = \psi(x) \\ u(0,t) = \mu_1(t) \qquad u(l,t) = \mu_2(t) \end{cases} \qquad \begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = f(x,t) \quad t>0 \\ u|_{t=0} = \varphi(x) \qquad \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = \psi(x) \\ u|_{x=0} = u|_{x=l} = 0 \end{cases}$$

$$\text{(P1)}\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = 0 \quad t>0 \\ u|_{t=0} = \varphi(x) \qquad \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = \psi(x) \\ u|_{x=0} = u|_{x=l} = 0 \end{cases} \qquad (P_2) \begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = f(x,t) \quad t>0 \\ u|_{t=0} = \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = 0 \\ u|_{x=0} = u|_{x=l} = 0 \end{cases}$$

(P1)Fourier 方法：$u(x,t) = X(x)T(t) \Rightarrow T''(t)X(x) - a^2T(t)X''(x) = 0 \Rightarrow \dfrac{X''(x)}{X(x)} = \dfrac{T''(t)}{a^2T(t)} \triangleq -\mu$。于是 $\mu_k = \left(\dfrac{k\pi}{l}\right)^2 \quad k = 1,2,\ldots, \quad X_k(x) = C_k\sin\dfrac{k\pi}{l}x \quad k = 1,2,\ldots. T''(t) + \left(\dfrac{k\pi}{l}\right)^2 a^2T(t) = 0 \Rightarrow$
$T_k(t) = A_k\cos\dfrac{ak\pi}{l}t + B_k\sin\dfrac{ak\pi}{l}t \quad k = 1,2,\ldots$（注意上面是一阶导，这里是二阶导。）

$$u(x,t) = \sum_{k=1}^{\infty} u_k(x,t) = \sum_{k=1}^{\infty}\left(A_k\cos\frac{ak\pi}{l}t\sin\frac{k\pi}{l}x + B_k\sin\frac{ak\pi}{l}t\sin\frac{k\pi}{l}x\right)$$

$$\begin{cases}\varphi(x)=\sum\limits_{k=1}^{\infty}A_k\sin\dfrac{k\pi}{l}x\\ \psi(x)=\sum\limits_{k=1}^{\infty}B_k\dfrac{k\pi a}{l}\sin\dfrac{k\pi}{l}x\end{cases}\xrightarrow{\text{正交系}\{\sin\frac{k\pi}{l}x\}}\begin{cases}A_k=\dfrac{2}{l}\displaystyle\int_0^l\varphi(x)\sin\dfrac{k\pi}{l}x\,dx\\ B_k=\dfrac{2}{ak\pi}\displaystyle\int_0^l\psi(x)\sin\dfrac{k\pi}{l}x\,dx\end{cases}$$

对于 (P_2)，$w(x,t,\tau)$ 为 $\begin{cases}\dfrac{\partial^2 w}{\partial t^2}-a^2\dfrac{\partial^2 w}{\partial x^2}=0\quad(t>\tau)\\ w|_{t=\tau}=0\qquad \left.\dfrac{\partial w}{\partial t}\right|_{t=\tau}=f(x,\tau)\\ w(0,t)=w(l,t)=0\end{cases}$ 的解。

$$w(x,t,\tau)=\sum_{k=1}^{\infty}B_k(\tau)\sin\frac{ak\pi}{l}(t-\tau)\sin\frac{k\pi}{l}x,\quad B_k(\tau)=\frac{2}{k\pi a}\int_0^l f(x,\tau)\sin\frac{k\pi}{l}x\,dx$$

$u_2(x,t)=\int_0^t\sum_{k=1}^{\infty}B_k(\tau)\sin\dfrac{ak\pi}{l}(t-\tau)\sin\dfrac{k\pi}{l}x\,d\tau$。

6、一端固定的半无界弦振动问题 $\begin{cases}\dfrac{\partial^2 u}{\partial t^2}-a^2\dfrac{\partial^2 u}{\partial x^2}=0\quad(t>0,\ x>0)\\ u|_{t=0}=\varphi(x)\qquad \left.\dfrac{\partial u}{\partial t}\right|_{t=0}=\psi(x)\quad(x>0)\\ u(0,t)=0\end{cases}$ 。

考虑在 $x<0$ 仍有弦存在，而振动过程保持 x=0 不动。记 $\Phi(x)$, $\Psi(x)$ 分别为初值 $\varphi(x),\psi(x)$ 延拓后的函数。由达朗贝尔公式 $u(x,t)=\dfrac{1}{2}(\Phi(x+at)+\Phi(x-at))+\dfrac{1}{2a}\displaystyle\int_{x-at}^{x+at}\Psi(y)\,dy$。

$u(0,t)=0\Rightarrow u(0,t)=\dfrac{1}{2}(\Phi(at)+\Phi(-at))+\dfrac{1}{2a}\displaystyle\int_{-at}^{at}\Psi(y)\,dy=0$，当 $\Phi(x),\Psi(x)$ 是奇函数时一定成立，从而将 $\varphi(x),\psi(x)$ 奇延拓得到 $\Phi(x),\Psi(x)$。于是代回公式：

$$\begin{cases}x-at>0\text{时}(x+at>0),u(x,t)=\dfrac{1}{2}(\varphi(x-at)+\varphi(x+at))+\dfrac{1}{2a}\displaystyle\int_{x-at}^{x+at}\psi(y)\,dy\\ x-at<0\text{但}x+at>0\text{ 时},u(x,t)=\dfrac{1}{2}(\varphi(x+at)-\varphi(at-x))+\dfrac{1}{2a}\displaystyle\int_{at-x}^{x+at}\psi(y)\,dy\end{cases}$$

7、例：用波的传播分析建立达朗贝尔公式与 Fourier 方法得到的解之间的关系。
实际上就是把两个函数 Fourier 展开，然后表出由 Fourier 方法得到的解。

8、n=3 的波动方程的 Cauchy 问题。$\begin{cases}\dfrac{\partial^2 u}{\partial t^2}-a^2\Delta u=f(x,t)\quad t>0\\ u|_{t=0}=g_0(x)\\ \left.\dfrac{\partial u}{\partial t}\right|_{t=0}=g_1(x)\end{cases}$ $f\in C^2,\ g_0(x)\in C^3,\ g_1(x)\in C^2$ 。

设 $\tilde u(x,t)=H(t)u(x,t)\overset{t\geq0}{=}u(x,t)$，$\left(\dfrac{\partial^2}{\partial t^2}-a^2\Delta\right)\tilde u(x,t)=F(x,t)\Rightarrow\tilde u(x,t)=E_+(x,t)*F(x,t)\overset{t\geq0}{=}u(x,t)$。$F(x,t)=\delta'(t)u(x,t)+2\delta(t)\dfrac{\partial u}{\partial t}+H(t)f(x,t)$。$\delta'(t)u(x,t)=-\delta(t)\dfrac{\partial u}{\partial t}+\dfrac{\partial}{\partial t}\Big(\delta(t)u(x,t)\Big)$。

于是$\dfrac{\partial^2\tilde u}{\partial t^2}-a^2\Delta\tilde u=-\delta(t)\dfrac{\partial u}{\partial t}+\dfrac{\partial}{\partial t}(\delta(t)u(x,t))+H(t)f(x,t)=H(t)f(x,t)+\delta(t)g_1(x)+\delta'(t)g_0(x)$

下面分别对三个部分做卷积，求结果。其中 $E_+(x,t)=\dfrac{H(t)}{4\pi a^2t}\delta(at-|x|)$.

$$\begin{aligned}\langle E_+(x,t)*(\delta(t)g_1(x)),\varphi(x,t)\rangle&=\langle E_+(x,t),\langle\delta(\tau)g_1(y),\varphi(x+y,t+\tau)\rangle\rangle\\&=\langle E_+(x,t),\langle g_1(y),\varphi(x+y,t)\rangle\rangle=E_+(x,t)\underset{(x)}{*}g_1(x)\end{aligned}$$

$$E_+(x,t)*\delta'(t)g_0(x)=E_+(x,t)*\frac{\partial}{\partial t}(\delta(t)g_0(x))=\frac{\partial}{\partial t}(E_+(x,t)\underset{(x)}{*}g_0(x))$$

于是 $\tilde{u}(x,t)=H(t)u(x,t)=E_+(x,t)*(H(t)f(x,t))+\frac{\partial}{\partial t}(E_+\underset{(x)}{*}g_0(x))+E_+\underset{(x)}{*}g_1(x)=I_1+I_2+I_3$。

$$\begin{aligned}I_1=E_+(x,t)*(H(t)f(x,t))&=\int_{\mathbb{R}}\int_{\mathbb{R}^3}H(\tau)f(y,\tau)H(t-\tau)\frac{1}{4\pi a^2(t-\tau)}\delta(a(t-\tau)-|x-y|)\,dy\,d\tau\\&=\int_0^t\mathrm{d}\tau\int_{|x-y|=a(t-\tau)}\frac{f(y,\tau)}{4\pi a^2(t-\tau)}\mathrm{d}y=\frac{1}{4\pi a}\int_0^t\mathrm{d}\tau\int_{|x-y|=a(t-\tau)}\frac{f(y,\tau)}{|x-y|}\mathrm{d}S_y\\&=\frac{1}{4\pi a^2}\int_{|x-y|\leqslant at}\frac{f\left(y,t-\frac{|x-y|}{a}\right)}{|x-y|}\mathrm{d}y\end{aligned}$$

$$I_3=E_+(x,t)\underset{(x)}{*}g_1(x)=\frac{H(t)}{4\pi a^2t}\int_{\mathbb{R}^3}g_1(y)\delta(at-|x-y|)\mathrm{d}y=\frac{1}{4\pi a^2t}\int_{|x-y|=at}g_1(y)\mathrm{d}S_y=tM\{g_1\}$$

同理, $I_2=\partial_t(tM\{g_0\}).\tilde{u}(x,t)=\frac{1}{4\pi a^2}\int_{|x-y|\leqslant at}\frac{f\left(y,t-\frac{|x-y|}{a}\right)}{|x-y|}\mathrm{d}y+\partial_t(tM\{g_0\})+tM\{g_1\}\overset{t>0}{=\!=}u(x,t)$。

9、降维法。二维。$\begin{cases}\left(\frac{\partial}{\partial t}-a^2\left(\frac{\partial^2}{\partial x_1^2}+\frac{\partial^2}{\partial x_2^2}\right)\right)u=0\\ u|_{t=0}=g_0(x),\quad \left.\frac{\partial u}{\partial t}\right|_{t=0}=g_1(x)\end{cases}$ $.u(x,t)=\partial_t(tM(g_0))+tM(g_1)$。$tM(g_1)=\frac{1}{4\pi a^2t}\int_{|x-y|=at}g_1(y)\mathrm{d}S_y,\quad \cos\theta\mathrm{d}S_y=\mathrm{d}y_1\mathrm{d}y_2,\cos\theta=\frac{\sqrt{(at)^2-((x_1-y_1)^2+(x_2-y_2)^2)}}{at}$。

$$tM(g_1)=\frac{1}{2\pi a}\int_{\sqrt{(x_1-y_1)^2+(x_2-y_2)^2}\leqslant at}\frac{g_1(y_1,y_2)\mathrm{d}y_1\mathrm{d}y_2}{\sqrt{(at)^2-((x_1-y_1)^2+(x_2-y_2)^2)}}.$$

$$\partial_t(tM(g_0))=\frac{1}{2\pi a}\frac{\partial}{\partial t}\int_{\sqrt{(x_1-y_1)^2+(x_2-y_2)^2}\leqslant at}\frac{g_0(y_1,y_2)\mathrm{d}y_1\mathrm{d}y_2}{\sqrt{(at)^2-((x_1-y_1)^2+(x_2-y_2)^2)}}.$$

10、降维法。一维。$\begin{cases}\left(\frac{\partial^2}{\partial t^2}-a^2\frac{\partial^2}{\partial x^2}\right)u=0\\ u|_{t=0}=g_0(x),\quad \left.\frac{\partial u}{\partial t}\right|_{t=0}=g_1(x)\end{cases}$ $.u(x,t)=\partial_t(tM(g_0))+tM(g_1)$。

$$\frac{2\pi r\mathrm{d}y_1}{\mathrm{d}S}=\cos\theta=\frac{r}{at}\Rightarrow\mathrm{d}S=\frac{at2\pi r\mathrm{d}y_1}{r}=2\pi at\mathrm{d}y_1$$

$$tM(g_1)=\frac{1}{4\pi a^2t}\int_{|x-y|=at}g_1(y)\mathrm{d}S_y=\frac{1}{4\pi a^2t}\int_{x_1-at}^{x_1+at}g_1(y_1)2\pi at\mathrm{d}y_1=\frac{1}{2a}\int_{x_1-at}^{x_1+at}g_1(y_1)\mathrm{d}y_1$$

$$\partial_t(tM(g_0))=\partial_t\left(\frac{1}{2a}\int_{x_1-at}^{x_1+at}g_0(y_1)\mathrm{d}y_1\right)=\frac{1}{2a}(g_0(x_1+at)a-g_0(x_1-at)(-a))=\frac{1}{2}(g_0(x_1+at)+g_0(x_1-at))$$

于是 $u(x,t)=\frac{1}{2a}\int_{x_1-at}^{x_1+at}g_1(y_1)\mathrm{d}y_1+\frac{1}{2}(g_0(x_1+at)+g_0(x_1-at))$ 即为一维齐次情形的 d'Alembert 解。

11、证明函数 $E(x,t)=\begin{cases}\frac{1}{2a}, & |x|\leqslant at,t\geqslant 0\\ 0, & \text{if not}\end{cases}$ 是弦振动方程的基本解。

即证 $\left\langle\left(\dfrac{\partial^2}{\partial t^2}-a^2\dfrac{\partial^2}{\partial x^2}\right)E(x,t),\varphi(x,t)\right\rangle=\langle\delta(x,t),\varphi(x,t)\rangle=\varphi(0,0),\ \forall\varphi(x,t)\in C_0^\infty$。

$\left\langle\left(\dfrac{\partial^2}{\partial t^2}-a^2\dfrac{\partial^2}{\partial x^2}\right)E(x,t),\varphi(x,t)\right\rangle=\displaystyle\int_{x^2\leqslant(at)^2,t\geqslant0}\frac{1}{2a}\left(\frac{\partial^2}{\partial t^2}-a^2\frac{\partial^2}{\partial x^2}\right)\varphi(x,t)\,\mathrm{d}x\mathrm{d}t(*)$. 令 $y=x+at$, $z=at-x$，被积函数变换：$\left(\dfrac{\partial^2}{\partial t^2}-a^2\dfrac{\partial^2}{\partial x^2}\right)\varphi=4a^2\dfrac{\partial^2\varphi}{\partial y\partial z},\quad J(y,z)=\begin{vmatrix}\dfrac{\partial x}{\partial y}&\dfrac{\partial x}{\partial z}\\\dfrac{\partial t}{\partial y}&\dfrac{\partial t}{\partial z}\end{vmatrix}=\dfrac{1}{2a}.$

被积区域变换：$\left.\begin{matrix}x^2\leqslant(at)^2\Rightarrow y,z\geqslant0\\t\geqslant0\Rightarrow y+z\geqslant0\end{matrix}\right\}\Rightarrow y\geqslant0,z\geqslant0.$

$$(*)=\int_{y=0}^{+\infty}\int_{z=0}^{+\infty}\frac{1}{2a}4a^2\frac{\partial^2\varphi}{\partial y\partial z}\frac{1}{2a}\,\mathrm{d}y\mathrm{d}z\xlongequal{\varphi\in C_0^\infty}\int_{y=0}^{+\infty}\int_{z=0}^{+\infty}\frac{\partial}{\partial z}\left(\frac{\partial\varphi}{\partial y}\right)\mathrm{d}z\mathrm{d}y\xrightarrow[\varphi\in C_0^\infty]{\text{分部积分}}(-1)(-1)\varphi(0,0)=\varphi(0,0).$$

12、齐次方程齐次边初值条件能量守恒。动能 $U=\dfrac12\displaystyle\int_\Omega\rho u_t^2\,\mathrm{d}x\mathrm{d}y$，位能 $V=\dfrac12\displaystyle\int_\Omega T(u_x^2+u_y^2)\,\mathrm{d}x\mathrm{d}y$。

$$E(t)=\int_\Omega u_t^2+a^2(u_x^2+u_y^2)\,\mathrm{d}x\mathrm{d}y,\quad a^2=\frac{T}{\rho}.$$

对 $\begin{cases}\left(\dfrac{\partial^2}{\partial t^2}-a^2\left(\dfrac{\partial^2}{\partial x^2}+\dfrac{\partial^2}{\partial y^2}\right)\right)u=0\\u|_{t=0}=\varphi(x,y),\quad\left.\dfrac{\partial u}{\partial t}\right|_{t=0}=\psi(x,y),\quad u|_{\partial\Omega}=0\end{cases}$ 证明: $\dfrac{\mathrm{d}E(t)}{\mathrm{d}t}=0$。

对 $\dfrac{\mathrm{d}E}{\mathrm{d}t}=\displaystyle\int_\Omega 2u_tu_{tt}+a^2(2u_xu_{xt}+2u_yu_{yt})\,\mathrm{d}x\mathrm{d}y$ 凑微分 $u_xu_{xt}=\partial_x(u_xu_t)-u_{xx}u_t;u_yu_{yt}=\partial_y(u_yu_t)-u_{yy}u_t$。

$$\begin{aligned}\frac{\mathrm{d}E}{\mathrm{d}t}&=2\int_\Omega u_tu_{tt}+a^2(\partial_x(u_xu_t)-u_{xx}u_t+\partial_y(u_yu_t)-u_{yy}u_t)\,\mathrm{d}x\mathrm{d}y\\&=2\int_\Omega u_t\big(u_{tt}-a^2(u_{xx}+u_{yy})\big)\mathrm{d}x\mathrm{d}y+2a^2\int_\Omega\partial_x(u_xu_t)+\partial_y(u_yu_t)\,\mathrm{d}x\mathrm{d}y\\&=2a^2\int_{\partial\Omega}u_xu_t\cos(\vec n,x)+u_yu_t\cos(\vec n,y)\,\mathrm{d}S=2a^2\int_{\partial\Omega}u_t\Big(u_x\cos(\vec n,x)+u_y\cos(\vec n,y)\Big)\mathrm{d}S=0\end{aligned}$$

13、初边值问题的唯一性。$\begin{cases}u_{tt}-a^2(u_{xx}+u_{yy})=f(x,y,t),\quad(x,y)\in\Omega\\u|_{t=0}=\varphi(x,y),\quad\partial_tu|_{t=0}=\psi(x,y),\quad u|_{\partial\Omega}=0\end{cases}$。

$u=u_1-u_2$ 能量守恒。$E(t)=E(0)=\displaystyle\int_\Omega\big(u_t^2+a^2(u_x^2+u_y^2)\big)\,\mathrm{d}x\mathrm{d}y\Big|_{t=0}$. 由 $u|_{t=0}=0$，$u_x|_{t=0}=0,u_y|_{t=0}=0\Rightarrow u_t^2+a^2(u_x^2+u_y^2)|_{t=0}=0\Rightarrow E(0)=0\Rightarrow u_t=u_x=u_y=0\Rightarrow u\equiv const$。这个常数为 0。

14、稳定性。$\begin{cases}u_{tt}-a^2(u_{xx}+u_{yy})=f(x,y,t)\qquad x\in\Omega,\\u|_{t=0}=\varphi,\partial_tu|_{t=0}=\psi,\quad u|_{\partial\Omega}=0.\end{cases}$. 对 $\forall\varepsilon>0,\exists\eta(\varepsilon,T)>0$, s.t. 当:

$\begin{cases}\|\varphi_1-\varphi_2\|_{L^2(\Omega)}<\eta\\\|\varphi_{1x}-\varphi_{2x}\|_{L^2(\Omega)}<\eta\\\|\varphi_{1y}-\varphi_{2y}\|_{L^2(\Omega)}<\eta\\\|\psi_1-\psi_2\|_{L^2(\Omega)}<\eta\\\|f_1-f_2\|_{L^2((0,T)\times\Omega)}<\eta\end{cases}$ 则 u_1 和 u_2 在 [0, T] 上成立 $\begin{cases}\|u_1-u_2\|_{L^2(\Omega)}<\varepsilon\\\|u_{1x}-u_{2x}\|_{L^2(\Omega)}<\varepsilon\\\|u_{1y}-u_{2y}\|_{L^2(\Omega)}<\varepsilon\\\|u_{1t}-u_{2t}\|_{L^2(\Omega)}<\varepsilon\end{cases}$.

由于可以和上面一样把后面一项化成在边界上后得 0，于是：

$$\frac{\mathrm{d}E}{\mathrm{d}t}=2\int_\Omega u_tu_{tt}+a^2(u_xu_{xt}+u_yu_{yt})\mathrm{d}x\mathrm{d}y=2\int_\Omega u_t\cdot f\mathrm{d}x\mathrm{d}y\leqslant\int_\Omega u_t^2+f^2\mathrm{d}x\mathrm{d}y\leqslant E(t)+\int_\Omega f^2\mathrm{d}x\mathrm{d}y$$

凑微分 $\dfrac{\mathrm{d}}{\mathrm{d}t}(e^{-t}E(t))\leqslant e^{-t}\displaystyle\int_\Omega f^2\mathrm{d}x\mathrm{d}y$。由于时间有限，对 E 做估计：

$$E(t)\leqslant e^t\left(E(0)+\int_0^t\int_\Omega e^{-\tau}f^2\mathrm{d}x\mathrm{d}y\mathrm{d}\tau\right)\Rightarrow E(t)\leqslant C_0\left(E(0)+\int_0^T\int_\Omega f^2\mathrm{d}x\mathrm{d}y\mathrm{d}t\right).$$

设 $E(t)\leqslant C_0\left(E(0)+\displaystyle\int_0^T\int_\Omega f^2\mathrm{d}x\mathrm{d}y\mathrm{d}t\right)$，则 $\dfrac{\mathrm{d}E_0(t)}{\mathrm{d}t}=\displaystyle\int_\Omega 2uu_t\mathrm{d}x\mathrm{d}y\leqslant\int_\Omega u_t^2+u^2\mathrm{d}x\mathrm{d}y\leqslant E_0(t)+E(t)$。

$$\int_\Omega u^2\mathrm{d}x\mathrm{d}y=E_0(t)\leqslant C_1\left(E_0(0)+E(0)+\int_0^T\int_\Omega f^2\mathrm{d}x\mathrm{d}y\mathrm{d}t\right).$$

设 $u=u_1-u_2$，则 u 满足：$\begin{cases}u_{tt}-a^2(u_{xx}+u_{yy})=f_1-f_2\quad \text{in }\Omega,\\ u|_{t=0}=\varphi_1-\varphi_2,\quad \partial_tu|_{t=0}=\psi_1-\psi_2,\quad u|_{\partial\Omega}=0.\end{cases}$

$$\begin{aligned}\|u_1-u_2\|^2_{L^2(\Omega)}&=E_0(t)\leqslant C_1\left(E_0(0)+E(0)+\int_0^T\int_\Omega(f_1-f_2)^2\mathrm{d}x\mathrm{d}y\mathrm{d}t\right)\\&=C_1\left(\left(\left.\int_\Omega u^2\mathrm{d}x\mathrm{d}y\right|_{t=0}+\left.\int_\Omega\left[u_t^2+a^2(u_x^2+u_y^2)\right]\mathrm{d}x\mathrm{d}y\right|_{t=0}\right)+C_1\|f_1-f_2\|^2_{L^2([0,T]\times\Omega)}\right)\\&\leqslant C_1\left(\|\varphi_1-\varphi_2\|^2_{L^2(\Omega)}+\|\psi_1-\psi_2\|^2+a^2\left(\|\varphi_{1x}-\varphi_{2x}\|^2+\|\varphi_{1y}-\varphi_{2y}\|^2\right)+\|f_1-f_2\|^2\right)\\&\leqslant(3+2a^2)C_1\eta^2\leqslant\tilde{C}\cdot\eta^2<\epsilon^2\end{aligned}$$

其余三项 $\|u_{1t}-u_{2t}\|^2_{L^2(\Omega)}<\varepsilon,\|u_{1x}-u_{2x}\|^2_{L^2(\Omega)}<\varepsilon,\|u_{1y}-u_{2y}\|^2_{L^2(\Omega)}<\varepsilon$ 只需用 $\displaystyle\int_\Omega u_t^2+a^2(u_x^2+u_y^2)\mathrm{d}x\mathrm{d}y\leqslant C_0\left(E(0)+\int_0^T\int_\Omega f^2\mathrm{d}x\mathrm{d}y\mathrm{d}t\right)$。

15、例：受摩擦力作用且具有固定端点的有界弦振动 $\dfrac{\partial^2u}{\partial t^2}-a^2\dfrac{\partial^2u}{\partial x^2}+c\dfrac{\partial u}{\partial t}=0,\quad c>0$。证明其能量减少，且对非齐次方程 $\dfrac{\partial^2u}{\partial t^2}-a^2\dfrac{\partial^2u}{\partial x^2}+c\dfrac{\partial u}{\partial t}=f$ 的初边值问题解唯一。

$$E(t)=\int_a^b\left(2\frac{\partial u}{\partial t}\frac{\partial^2u}{\partial t^2}+2a^2\frac{\partial u}{\partial x}\frac{\partial^2u}{\partial x\partial t}\right)\mathrm{d}x=2\int_a^b-c\left(\frac{\partial u}{\partial t}\right)^2\mathrm{d}x+2a^2\left.\frac{\partial u}{\partial x}\frac{\partial u}{\partial t}\right|_a^b$$

由于弦两端固定，即 $u|_{x=a}=u|_{x=b}=0$，$\dfrac{\mathrm{d}E(t)}{\mathrm{d}t}=-2c\displaystyle\int_a^b\left(\frac{\partial u}{\partial t}\right)^2\mathrm{d}x\leqslant0$。对边初值问题：

$$\begin{cases}\dfrac{\partial^2u}{\partial t^2}-a^2\dfrac{\partial^2u}{\partial x^2}+c\dfrac{\partial u}{\partial t}=f,\quad t>0\\ u|_{x=a}=0,\quad u|_{x=b}=0\quad(u|_{x=a}=\mu_1(t),\quad u|_{x=b}=\mu_2(t))\qquad \text{设 } u=u_1-u_2\text{。}\\ u|_{t=0}=\varphi(x),\quad \left.\dfrac{\partial u}{\partial t}\right|_{t=0}=\psi(x)\end{cases}$$

$E(0)=\displaystyle\int_a^b\left.\left[\left(\frac{\partial u}{\partial t}\right)^2+a^2\left(\frac{\partial u}{\partial x}\right)^2\right]\right|_{t=0}\mathrm{d}x=0$, 从而 $E(t)\equiv0\Rightarrow u_t=u_x=u_y=0\Rightarrow u\equiv0$。

16、变系数方程的能量法。$E(t)=\dfrac{1}{2}\displaystyle\int_0^l\left[k(x)\left(\frac{\partial u}{\partial x}\right)^2+\rho(x)\left(\frac{\partial u}{\partial t}\right)^2+q(x)u^2\right]\mathrm{d}x$,$k(x)\geqslant k_0>0,q(x)\geqslant 0,\rho(x)\geqslant\rho_0>0$。证明：对于方程 $\begin{cases}\rho(x)\dfrac{\partial^2 u}{\partial t^2}-\dfrac{\partial}{\partial x}\left(k(x)\dfrac{\partial u}{\partial x}\right)+q(x)u=0, & x\in(0,l),\quad t>0\\ u|_{t=0}=0, & \left.\dfrac{\partial u}{\partial t}\right|_{t=0}=0\\ u|_{x=0}=u|_{x=l}=0\end{cases}$ 解 $u\equiv 0$。

$$
\begin{aligned}
\frac{\mathrm{d}E(t)}{\mathrm{d}t}&=\frac{1}{2}\int_0^l\left[2k(x)\frac{\partial u}{\partial x}\frac{\partial^2 u}{\partial x\partial t}+2\rho(x)\frac{\partial u}{\partial t}\frac{\partial^2 u}{\partial t^2}+2q(x)u\frac{\partial u}{\partial t}\right]\mathrm{d}x\\
&=\int_0^l\left[-\frac{\partial}{\partial x}\left(k(x)\frac{\partial u}{\partial x}\right)\frac{\partial u}{\partial t}+\rho(x)\frac{\partial u}{\partial t}\frac{\partial^2 u}{\partial t^2}+q(x)u\frac{\partial u}{\partial t}\right]\mathrm{d}x+\left[k(x)\frac{\partial u}{\partial x}\frac{\partial u}{\partial t}\right]_0^l=0
\end{aligned}
$$

$E(t)=E(0)\equiv 0$，即得。

17、带 Robin 边界条件的二维波动方程。$\begin{cases}\dfrac{\partial^2 u}{\partial t^2}-a^2\left(\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}\right)=0, & x\in\Omega,\quad a^2=\dfrac{T}{\rho}\\ \left.\left(T\dfrac{\partial u}{\partial n}+\sigma u\right)\right|_{\partial\Omega}=0, & T>0,\quad \sigma>0\\ u|_{t=0}=0,\quad \partial_t u|_{t=0}=0\end{cases}$

$E(t)=\displaystyle\int_\Omega\left[\frac{\rho}{2}\left(\frac{\partial u}{\partial t}\right)^2+\frac{T}{2}\left(\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2\right)\right]\mathrm{d}x\mathrm{d}y+\int_{\partial\Omega}\frac{\sigma}{2}u^2\mathrm{d}s$，$E_0(t)=\displaystyle\int_\Omega u^2\mathrm{d}x\mathrm{d}y$。证明能量不等式：

$$
E_0(t)\leqslant e^tE_0(0)+\frac{2}{\rho}(e^t-1)E(0).
$$

$$
\begin{aligned}
\frac{\mathrm{d}E(t)}{\mathrm{d}t}&=\int_\Omega\rho\frac{\partial u}{\partial t}\frac{\partial^2 u}{\partial t^2}+T\left(\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\frac{\partial u}{\partial t}\right)-\frac{\partial^2 u}{\partial x^2}\frac{\partial u}{\partial t}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\frac{\partial u}{\partial t}\right)-\frac{\partial^2 u}{\partial y^2}\frac{\partial u}{\partial t}\right)\mathrm{d}x\mathrm{d}y+\int_{\partial\Omega}\sigma u\frac{\partial u}{\partial t}\mathrm{d}s\\
&=T\int_{\partial\Omega}\left[\frac{\partial u}{\partial x}\frac{\partial u}{\partial t}\cos(x,n)+\frac{\partial u}{\partial y}\frac{\partial u}{\partial t}\cos(y,n)\right]\mathrm{d}s+\int_{\partial\Omega}\sigma u\frac{\partial u}{\partial t}\mathrm{d}s\\
&=\int_{\partial\Omega}T\frac{\partial u}{\partial t}\underbrace{\left(\frac{\partial u}{\partial x}\cos(x,n)+\frac{\partial u}{\partial y}\cos(y,n)\right)}_{\frac{\partial u}{\partial n}}\mathrm{d}s+\int_{\partial\Omega}\sigma u\frac{\partial u}{\partial t}\mathrm{d}s=\int_{\partial\Omega}\frac{\partial u}{\partial t}\left(T\frac{\partial u}{\partial n}+\sigma u\right)\mathrm{d}s=0
\end{aligned}
$$

于是 $E(t)=E(0),\quad t>0$。

$$
\begin{aligned}
\frac{\mathrm{d}E_0(t)}{\mathrm{d}t}&=2\int_\Omega u\frac{\partial u}{\partial t}\mathrm{d}x\mathrm{d}y\leqslant\int_\Omega u^2\mathrm{d}x\mathrm{d}y+\int_\Omega\left(\frac{\partial u}{\partial t}\right)^2\mathrm{d}x\mathrm{d}y\\
&\leqslant E_0(t)+\frac{2}{\rho}E(t)=E_0(t)+\frac{2}{\rho}E(0)
\end{aligned}
$$

两边同乘 e^{-t}，再在 $(0,t)$ 上积分即可得证。

18、Cauchy 问题 $\begin{cases}u_{tt}-a^2(u_{xx}+u_{yy})=0,t>0\\ u|_{t=0}=\varphi(x),\quad u_t|_{t=0}=\psi(x)\end{cases}$ 考虑光锥 $\Omega_t:(x-x_0)^2+(y-y_0)^2\leqslant a^2(t-t_0)^2,\quad \Omega_0:(x-x_0)^2+(y-y_0)^2\leqslant a^2t_0^2$。定义局部能量：$E_1(\Omega_t)=\displaystyle\int_{\Omega_t}\left(u_t^2+a^2(u_x^2+u_y^2)\right)dxdy$。证明

$E_1(\Omega_t) \leqslant E_1(\Omega_0)$。即 $\dfrac{dE_1(\Omega_t)}{dt} \leqslant 0, \quad 0 \leqslant t \leqslant t_0$。

$$
\begin{aligned}
\frac{dE_1(\Omega_t)}{dt} &= \int_0^{a(t_0-t)}\int_0^{2\pi r}\frac{\partial}{\partial t}\left(u_t^2+a^2(u_x^2+u_y^2)\right)dsdr - a\int_{\partial\Omega_t}(u_t^2+a^2(u_x^2+u_y^2))ds \\
&= 2a^2\int_0^{a(t_0-t)}\int_0^{2\pi r}\left[\frac{\partial}{\partial x}(u_xu_t)+\frac{\partial}{\partial y}(u_yu_t)\right]dsdr - a\int_{\partial\Omega_t}(u_t^2+a^2(u_x^2+u_y^2))ds \\
&= -a\int_{\partial\Omega_t}\left(u_t^2+a^2(u_x^2+u_y^2)-2au_xu_t\cos(n,x)-2au_yu_t\cos(n,y)\right)\mathrm{d}s \leqslant 0 \quad \Rightarrow \quad E_1(\Omega_t) \leqslant E_1(\Omega_0)
\end{aligned}
$$

因为 $u_t^2+a^2u_x^2+a^2u_y^2-2au_xu_t\cos(n,x)-2au_yu_t\cos(n,y) = (u_t\cos(n,x)-au_x)^2+(u_t\cos(n,y)-au_y)^2 \geqslant 0$。

19、Cauchy 问题唯一性显然。下证稳定性。只要 $\|\varphi_1-\varphi_2\|_{L^2(\Omega_0)} \leqslant \eta, \quad \|\varphi_{1x}-\varphi_{2x}\|_{L^2(\Omega_0)} \leqslant \eta, \quad \|\varphi_{1y}-\varphi_{2y}\|_{L^2(\Omega_0)} \leqslant \eta, \quad \|\psi_1-\psi_2\|_{L^2(\Omega_0)} \leqslant \eta$ 则对应 (φ_1,ψ_1) 的解 u_1 与对应 (φ_2,ψ_2) 的解 u_2，有 u_1-u_2 在 $0 \leqslant t \leqslant \dfrac{R}{a}$ 上满足：$\|u_1-u_2\|_{L^2(\Omega_t)} \leqslant \varepsilon, \quad \|u_{1t}-u_{2t}\|_{L^2(\Omega_t)} \leqslant \varepsilon, \quad \|u_{1x}-u_{2x}\|_{L^2(\Omega_t)} \leqslant \varepsilon, \quad \|u_{1y}-u_{2y}\|_{L^2(\Omega_t)} \leqslant \varepsilon.$
又在锥体 K 上成立 $\|u_1-u_2\|_{L^2(K)} = \sqrt{\iiint_K (u_1-u_2)^2\mathrm{d}x\mathrm{d}y\mathrm{d}t} \leqslant \varepsilon$。

记 $E_0(\Omega_t) = \int_{\Omega_t} u^2\mathrm{d}x\mathrm{d}y$，在 $0 \leqslant t \leqslant t_0$ 时，有 $E_0(\Omega_t) \leqslant C(E_1(\Omega_0)+E_0(\Omega_0))$.

$\dfrac{\mathrm{d}E_0(\Omega_t)}{\mathrm{d}t} = 2\int_{\Omega_t} uu_t\mathrm{d}x\mathrm{d}y - a\int_{\partial\Omega_t} u^2\mathrm{d}s \leqslant 2\int_{\Omega_t} uu_t\mathrm{d}x\mathrm{d}y \leqslant E_0(\Omega_t)+E_1(\Omega_t) \leqslant E_0(\Omega_t)+E_1(\Omega_0).$

$$
\begin{aligned}
\|u_1-u_2\|_{L^2(\Omega_t)}^2 &= \int_{\Omega_t} u^2\mathrm{d}x\mathrm{d}y = E_0(\Omega_t) \leqslant C(E_0(\Omega_0)+E_1(\Omega_0)) \\
&= C\left(\int_{\Omega_0} u|_{t=0}^2\mathrm{d}x\mathrm{d}y + \int_{\Omega_0}\left(u_t^2+a^2(u_x^2+u_y^2)\right)\Big|_{t=0}\mathrm{d}x\mathrm{d}y\right) \\
&= C\Bigg(\underbrace{\int_{\Omega_0}(\varphi_1-\varphi_2)^2\mathrm{d}x\mathrm{d}y}_{\text{由}\eta^2\text{ 控制}} + \underbrace{\int_{\Omega_0}\left((\psi_1-\psi_2)^2+a^2\left((\varphi_{1x}-\varphi_{2x})^2+(\varphi_{1y}-\varphi_{2y})^2\right)\right)\mathrm{d}x\mathrm{d}y}_{\text{由}\eta^2\text{ 控制}}\Bigg) \\
&\leqslant \tilde{C}\eta^2
\end{aligned}
$$